

Ebrahim Sharifi, Ph.D., P.Eng.

Ph.D. Industrial Engineer | Business and Supply Chain Analytics

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SUMMARY

Ph.D. industrial engineer with expertise in business analytics, predictive modeling, and supply chain optimization. Taught **800+ students** across data mining, predictive modeling, and quantitative methods at Cape Breton University, earning **4.5 to 4.8 / 5** ratings. Industry background in supply chain analysis and project management. Seeking roles in business analytics, data analytics, or supply chain analytics in Canada.

CORE SKILLS

Machine Learning	XGBoost, Random Forest, Decision Trees, Naive Bayes, KNN, Neural Networks (ANN), K-Means Clustering, Classification, Regression, Feature Engineering, Model Evaluation
Tools and Languages	Python (pandas, scikit-learn), R, SQL, Excel, Tableau, MATLAB, GAMS, CPLEX
Analytics Methods	Predictive Modeling, Data Mining, Statistical Analysis, Forecasting, Optimization, Simulation, Multi-Criteria Decision Analysis, Time-Series Analysis
Supply Chain and Operations	Demand Forecasting, Inventory Management, Supplier Evaluation, Network Design, Robust Optimization, Stochastic Programming, Fuzzy Logic
Business Domains	Supply Chain and Logistics, Healthcare Analytics, E-commerce and Retail, Telecommunications, Manufacturing, Project Management
Communication	Stakeholder Reporting, Executive Briefings, Teaching and Mentorship, Cross-Functional Collaboration, Public Speaking

PROFESSIONAL EXPERIENCE

Assistant Professor (Business Analytics), Cape Breton University, Shannon School of Business *Sep 2022 to Apr 2026*
Sydney, NS, Canada

- Designed and delivered four post-baccalaureate analytics courses (Data Mining, Data Visualization, Predictive Modeling, Quantitative Methods) across **6 sections/year** with 40–60 students each, plus 3 additional sections. Cumulative: **800+ students** over three years.
- Earned consistent **4.5 to 4.8 out of 5** on official student evaluations (Fall 2025, n=45, 85% response rate). Top dimension: returning assignments in a timely manner (**4.77 / 5**).
- Supervised **100+ final group projects** on real datasets (Kaggle, etc.) in healthcare, e-commerce, telecom. Coached teams through full ML pipelines: data cleaning, feature engineering, model training, evaluation, presentations using **XGBoost, Random Forest, Decision Trees, Naive Bayes, KNN, ANN, K-Means** in Python and R.
- Co-supervised graduate research; co-authored applied analytics work with **460+ citations** on Google Scholar (h-index 7).

Research Assistant (Operations and Decision Analytics), Toronto Metropolitan University (formerly Ryerson) *Sep 2020 to Nov 2023*
Toronto, ON, Canada

- Built **multi-objective optimization and predictive models** for sustainable supply chain decisions on wheat, soybean, and biomass case studies. Presented at **Canadian Operations Research Society (CORS), INFORMS, IISE** conferences and chaired Supply Chain Optimization session. Published in top-tier journals (IF up to 12.1).

Project Manager, Mehr Naghsh Afarinan Shayan Company *Sep 2017 to Jul 2020*
Tehran, Iran

- Led a team of 5 to develop a **tourism mobile application** end-to-end: scope, scheduling, resource allocation, and delivery.
- Managed **stakeholder engagement and fund acquisition**, presenting business cases to investors and coordinating across technical and business teams to keep the project on track.

- Applied **multi-criteria decision analysis (MCDA)** to evaluate replacement vehicle for import/assembly (technical, cost, market criteria) and conduct market analysis on bus procurement from China. Automated recurring reporting processes across strategic planning centre.

SELECTED PROJECTS

Sustainable Food Supply Chain Optimization Under Uncertainty

Two published multi-objective optimization studies on sustainable, resilient agri-food supply chains under uncertainty. For an Ontario wheat network, intermodal transport cut costs 8% and delivery times 20% versus truck-only while holding resilience under winter disruption. For the Canadian soybean export chain, a two-stage stochastic model balanced profit, jobs, sustainability, and CO₂ across 18 scenarios, with SDG incentives offsetting a 12% profit gap from sustainable sourcing. Published in Journal of Cleaner Production (IF 9.8) and Sustainable Production and Consumption (IF 12.1).

Methods and tools: Multi-objective and two-stage stochastic optimization, Fuzzy Robust Stochastic programming, Best-Worst Method (IT2TrF-BWM), augmented ϵ -constraint, IBM CPLEX

Advancing Minority Class Detection in Diabetic Diagnosis

ML pipeline for multiclass diabetes prediction (healthy/prediabetic/diabetic) using CDC BRFSS dataset (253,680 records, 21 features, 84.2% class imbalance). Evaluated 6 resampling strategies across 5 classifiers to address severe underrepresentation of diabetic and prediabetic classes. Applied Borda count consensus ranking for robust feature selection. Key result: Tomek Links paired with lightweight boosting (LightGBM, CatBoost) achieved best performance for early diabetes detection, with 15% improvement in recall for minority classes compared to baseline models. Emphasis on minimizing false negatives to support clinical decision-making.

Methods and tools: Optuna hyperparameter tuning, Borda count feature selection, SMOTE/Tomek Links resampling, LightGBM, CatBoost, Python

Conversion Pattern Mining for Digital Marketing Campaigns [\(GitHub\)](#)

End-to-end data mining project on 8,000 customer records identifying the behaviors and campaign traits that drive conversions. Ran a full KDD pipeline — exploratory analysis, three classifiers (Logistic Regression, Decision Tree, Random Forest), and an Apriori association-rule miner implemented from scratch. Addressed an imbalanced target (87.6% converted) with class weighting and evaluated on lift and ROC-AUC alongside accuracy. Key result: Random Forest reached 0.90 accuracy and 0.81 ROC-AUC, with behavioral engagement signals (time on site, pages per visit, click-through rate, ad spend) far outweighing demographics; Apriori confirmed Conversion-type campaigns and high ad spend converting at 95–96% confidence. Translated results into business recommendations on real-time lead scoring and budget reallocation.

Methods and tools: Python (pandas, scikit-learn), Logistic Regression, Decision Tree, Random Forest, Apriori (from scratch), matplotlib, seaborn

EDUCATION

Ph.D., Industrial Engineering

2020 to 2023

Toronto Metropolitan University, Toronto, ON. Grade: A+ (GPA 4.33 / 4.33). Completed in 3 years. Nominated for the Governor General's Gold Medal.

M.Sc., Industrial Engineering (Engineering Management)

2016 to 2018

Iran University of Science and Technology. GPA: 90 / 100.

CERTIFICATIONS

- **Data Science Fundamentals with Python and SQL** (5-course specialization), IBM, 2026
- **Advanced Business Analytics** (5-course specialization), University of Colorado Boulder, 2026
- **Professional Engineer (P.Eng.)**, Professional Engineers Ontario (PEO), 2026
- **Diploma in Operational research**, Canadian Operations Research Society (CORS), 2021

HONOURS AND AFFILIATIONS

- Nominated for **Governor General's Gold Medal** for outstanding dissertation (highest academic honour in Canada).
- TMU Doctoral Tuition Scholarship (3 years full tuition + \$24,000)
- Toronto Met Graduate Scholarship (\$15,000) and MIE Graduate Studies Award (\$2,000).
- **Professional Membership:** CORS, INFORMS, ASAC, PEO.